

ARKOO PREBUILD INTELLIGENCE

Complete Application Architecture, Module Functionality & RBAC Guide

AI-Powered Meeting Transcriptions, Multi-Stakeholder Task Workflow & Role-Based Access Control System

1. Executive Overview & System Architecture

Arkoo Prebuild Intelligence is a state-of-the-art enterprise resource management and intelligence platform. The application integrates real-time browser audio recording, Gemini / OpenAI AI transcriptions, automatic task extraction, and strict Administrator-Controlled Role-Based Access Control (RBAC).

Key Architectural Highlights:

- **Single-Page Application (React 19 + TypeScript + Vite + TailwindCSS)**
- **Backend Server & Express Middleware (tsx server.ts)**
- **Cloud Database: Google Firebase Firestore (Master Stakeholders, Meetings, Tasks, Settings, Audit Logs)**
- **Security & Authentication: Google OAuth2 via Firebase Auth with zero self-registration (Admin pre-approval required)**
- **Integration Services: Persistent Google Drive Cloud Backup, Groq & NVIDIA AI Inference, Gmail SMTP Notifications**

2. Complete Module & Tab-by-Tab Functionality

Tab 1: Dashboard (/dashboard)

[Visible to: Admin, Manager, Employee, Vendor, Supplier, Other]

Centralized operational overview displaying real-time metrics, system health, and task analytics.

- **Metrics Cards:** Total Tasks, Active Meetings, Registered Stakeholders, and Pending Requests.
- **Priority & Status Breakdown:** Visual charts categorizing tasks into Low, Medium, High, Critical priorities and Pending, In-Progress, Completed states.
- **Real-Time Activity Timeline:** Instant stream of system activities including newly created tasks, meeting uploads, and status changes.
- **Stakeholder Distribution Tally:** Breakdown of active users categorized into Employees, Managers, Vendors, and Suppliers.

Tab 2: Meetings (/meetings)

[Visible to: Admin, Manager, Employee, Vendor, Supplier]

Repository of processed audio meeting recordings, AI summaries, and automated action item extraction.

- **Audio Transcripts:** Full word-for-word text transcription generated via OpenAI Whisper & Gemini AI.
- **Key Takeaways & Summaries:** AI-generated executive summaries highlighting decisions made during meetings.
- **Automated Action Items:** AI automatically detects task commitments from spoken dialogue and prompts one-click task creation.
- **Search & Filtering:** Instant filter by meeting title, status (recording, processing, completed), or date range.

Tab 3: Record Meeting (/record-meeting)

[Visible to: Admin, Manager Only]

Interactive audio recording workspace for capturing live discussions or uploading pre-recorded audio files.

- **Browser Microphone Capture:** Real-time audio recording with live waveform frequency visualization.
- **Audio File Upload:** Supports uploading audio files (.mp3, .wav, .m4a, .webm) up to 100MB.
- **Background Processing Pipeline:** Automatic ingestion -> Speech-to-Text conversion -> Gemini Key Takeaways parsing -> Firestore sync.
- **Restricted Access:** Strictly hidden and blocked for non-management roles to ensure meeting confidentiality.

Tab 4: Tasks (/tasks)

[Visible to: Admin, Manager, Employee, Vendor, Supplier, Other]

Comprehensive multi-stakeholder task management system supporting Kanban and List view modes.

- **Kanban & List Views:** Drag-and-drop task workflow management across Pending, In-Progress, and Completed columns.
- **Rich Task Details:** Title, detailed description, priority (Low/Med/High/Critical), department, assigned employee/vendor, and due date.
- **Subtasks Checklist:** Breakdown of complex tasks into individual checkable sub-action items.
- **Voice Notes & Activity Threads:** Stakeholders can attach voice recordings, comments, and file attachments directly to tasks.
- **Comprehensive Filters:** Filter tasks by assigned stakeholder, priority, department, status, or due horizon (overdue, today, this week).

2. Complete Module Breakdown (Continued)

Tab 5: Stakeholders / Master Directory (/employees) <i>Master directory for managing internal personnel and external vendor/supplier relationships.</i> • Master Employee List: Full profiles including Official Email, Personal Email, Department, Phone, Blood Group, Emergency Contact. • Stakeholder Types: Categorization into Employee, Manager, Vendor, Supplier, or Other. • Approval Status Lifecycle: Pending Approval, Active, and Rejected states. • Admin Invitation System: Triggers automated SMTP invitation email with sign-in instructions upon approval. • Status Management: Admins can activate, deactivate, approve, or delete stakeholder records.	[Visible to: Admin, Manager, Employee]
Tab 6: Role Management (/role-management) <i>Granular Role-Based Access Control (RBAC) engine for defining custom roles and feature permissions.</i> • Permission Matrix: Toggle feature access across Dashboard, Meetings, Record Meeting, Tasks, Employees, Settings, and Recycle Bin. • Custom Role Creation: Create new organizational roles tailored to specific company hierarchies. • Two-Stage Deletion Guardrails: Stage 1 soft-delete warning and Stage 2 confirmation to prevent accidental loss of role definitions. • System Default Safeguards: Default roles (Admin, Manager, Employee, Vendor, Supplier) are locked to preserve system integrity.	[Visible to: Admin Only]
Tab 7: Settings (/settings) <i>System-wide administrative configuration center for external integrations and AI providers.</i> • Google Drive Integration: Connect persistent Google Drive OAuth credentials for cloud storage of meeting audio & task attachments. • AI Provider Configuration: Configure API keys and model parameters for Groq AI (Llama 3.3) and NVIDIA NIM endpoints. • Stakeholder Options: Add/remove custom stakeholder dropdown options and department tags. • Notification Preferences: Configure SMTP email notification rules and system alerts.	[Visible to: Admin Only]
Tab 8: Recycle Bin (/recycle-bin) <i>Safety buffer holding soft-deleted tasks, meetings, and stakeholder records.</i> • Restore Capability: Instantly restore soft-deleted items back to active status without data loss. • Permanent Purge: Admins can permanently delete items to clean up database storage. • Audit Tracking: Tracks deletion timestamp and performing user ID.	[Visible to: Admin Only]

3. Role-Based Access Control (RBAC) Master Matrix

The table below details exact feature permissions, tab visibility, and action capabilities for every assigned role in the system:

Feature / Action	Admin	Manager	Employee	Vendor	Supplier	Other
View Dashboard (/dashboard)	FULL	FULL	FULL	FULL	FULL	LIMITED
View Meetings List (/meetings)	FULL	FULL	FULL	SHARED	SHARED	NONE
Record & Process Audio Meeting	FULL	FULL	NONE	NONE	NONE	NONE
View Tasks List (/tasks)	ALL TASKS	ALL TASKS	ASSIGNED	ASSIGNED	ASSIGNED	ASSIGNED
Create & Assign Tasks	FULL	FULL	FULL	SELF/DEPT	SELF/DEPT	NONE
Delete Tasks & Voice Notes	FULL	FULL	OWN ONLY	NONE	NONE	NONE
Add Task Voice Notes & Comments	FULL	FULL	FULL	FULL	FULL	NONE
View Stakeholders Directory	FULL	FULL	FULL	NONE	NONE	NONE
Add / Edit Stakeholder Profile	FULL	FULL	SELF ONLY	NONE	NONE	NONE
Approve / Reject Stakeholder	FULL	FULL	NONE	NONE	NONE	NONE
Manage System Settings	FULL	NONE	NONE	NONE	NONE	NONE
Manage Roles & Permissions	FULL	NONE	NONE	NONE	NONE	NONE
Recycle Bin Recovery & Purge	FULL	NONE	NONE	NONE	NONE	NONE

4. Security Rules & Self-Registration Governance

- Self-Registration Disabled: Public sign-ups are strictly forbidden. Users attempting to sign in with Google whose email addresses are not pre-added to the Firestore Employees Master Directory will be rejected immediately with an Access Denied message.
- System Super Admin Override: Hardcoded dedicated addresses (e.g., admin@arkooprebuild.com, anushakanna19@gmail.com) bypass directory verification to guarantee emergency administrative recovery.
- Stakeholder Approval Workflow: When a new stakeholder is added, their status defaults to 'Pending' or 'Active'. Pending users cannot access any app modules until an Admin or Manager sets their status to 'Active'.
- Firestore Security Rules Enforcement: All read/write operations on meetings, tasks, and settings are validated server-side by firestore.rules to prevent client-side authorization bypass.